

$$C(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n \sqrt{1-\zeta^2} t + \cos^{-1} \zeta)$$

$$\frac{dC(t)}{dt} = 0$$

After solving, General solution

$$t_p = \frac{n\pi}{\omega_n \sqrt{1-\zeta^2}} ; n=0, 1, 2, \dots$$

When, $n=1$ (1st overshoot) , When $n=2$ $\rightarrow$ (1st undershoot)

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$t_2 = \frac{2\pi}{\omega_n \sqrt{1-\zeta^2}}$$

4. Max peak overshoot (M_p):-

$$M_p = C(t_p) - 1$$

$$= \frac{e^{-\pi\zeta/\sqrt{1-\zeta^2}}}{\sqrt{1-\zeta^2}} e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

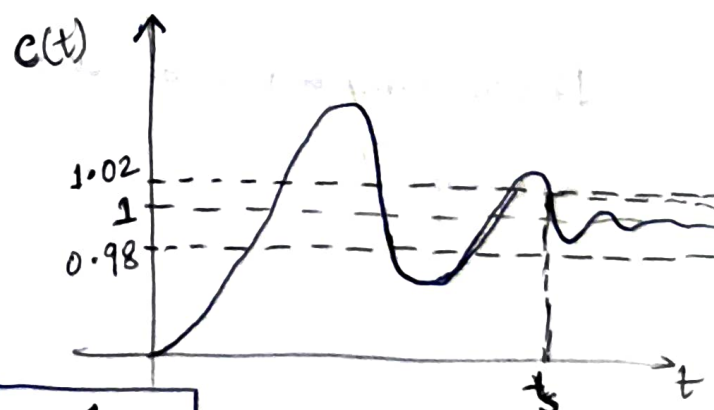
$$\% \text{ Maximum peak overshoot} = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$$

5. Settling time (t_s):-

$$1 \pm \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} = 1 \pm 2\%$$

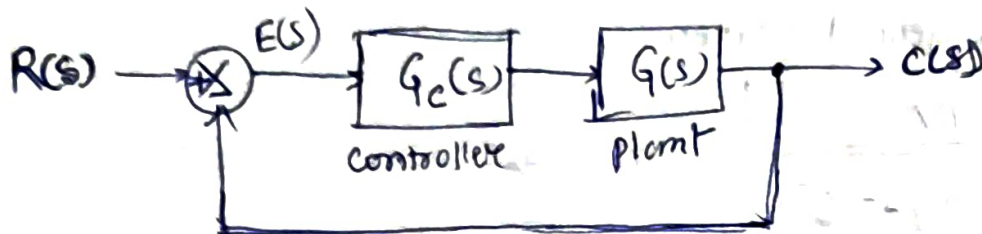
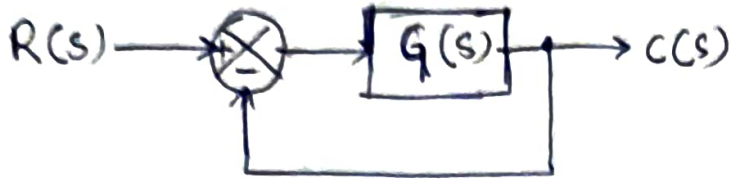
$$\frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} = 0.02$$

$$\Rightarrow e^{-\zeta \omega_n t} = 0.02 \Rightarrow t_s = \frac{4}{\zeta \omega_n}$$

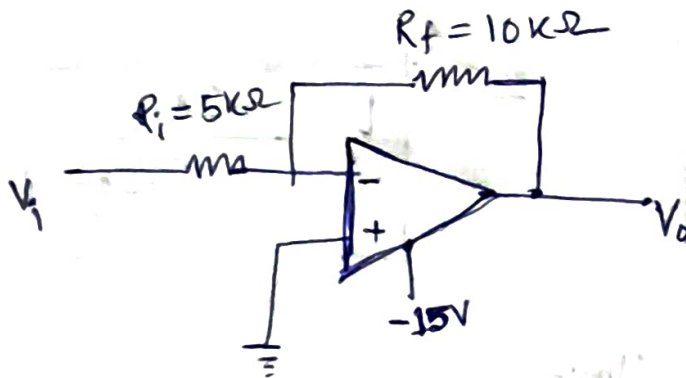


* Steady state error:-

$$E(s) = R(s) - C(s)$$



Assume that G_c uses an amplifier,

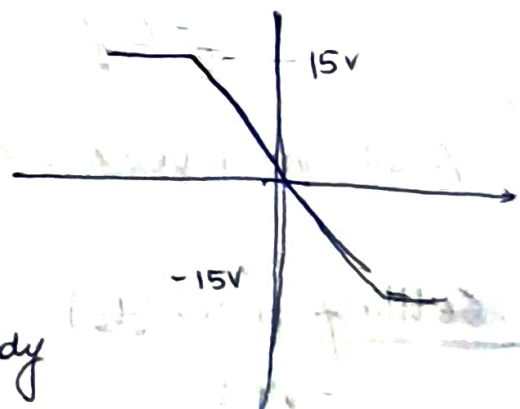


$$V_o = -\frac{R_f}{R_i} V_i$$

$$= -\frac{10}{5} V_i$$

$$= -2V_i$$

V_i	V_o
1	-2
2	-4
3	-6
4	-8
5	-10
10	-15



Non linearity of the controller causes steady state error

controlled q/p, $C(s) = \frac{G(s)}{1+G(s)} R(s)$

$$C(s) = E(s) G(s)$$

$$\therefore G(s) E(s) = \frac{G(s)}{1+G(s)} R(s)$$

$$E(s) = \frac{R(s)}{1+G(s)}$$

* Steady state error:-

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s), \text{ where } E(s) = \mathcal{L}\{e(t)\}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)}$$

, for linear system, steady state error depends on loop gain

* Unit step input:-

$$r(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1+G(s)} = \frac{1}{1+G(0)}$$

$$= \frac{1}{1+k_p}$$

where

$k_p = \lim_{s \rightarrow 0} G(s) \rightarrow$ position error const.

* Ramp i/p

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} s R(s) \frac{1}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + s G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s G(s)} = \frac{1}{K_v}$$

where, $K_v = \lim_{s \rightarrow 0} s G(s)$

Velocity error const.

* parabolic i/p

$$r(t) = \begin{cases} \frac{t^2}{2}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{2s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} s R(s) \frac{1}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}$$

$$= \frac{1}{K_a}$$

where

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)$$

= Acceleration error const.

* Consider the -2TF

$$G(s) = \frac{K (T_{z1}s + 1) (T_{z2}s + 1) \dots}{s^p (T_{p1}s + 1) (T_{p2}s + 1) \dots} \quad \text{(Time-constant form)}$$

type no. $\rightarrow$ No. of poles at origin
 $= p$

$$= \frac{K' (s + T_1) (s + T_2) \dots}{s^p (s + P_1) (s + P_2) \dots} \quad \text{(pole-zero form)}$$

where $K' = K \frac{\pi_i T_{zi}}{\pi_j T_{pj}}$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}, \text{ steady state error is dominated by the no. of poles at origin.}$$

Type-0 sys:-, $n=0$

Unit step

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$U(s) = \frac{1}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1 + G(s)}$$

$$= \frac{1}{1 + K}$$

Ramp i/p

$$R(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \cdot \frac{1}{s^2}}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + sG(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} sG(s)} = \frac{1}{K_v} = \infty$$

* Parabolic i/p

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{1}{s^3}}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)}$$

$$= \infty$$

Type-n

$$G(s) = \frac{K (T_{z1}s+1)(T_{z2}s+1) \dots}{s^n (T_{p1}s+1)(T_{p2}s+1) \dots}$$

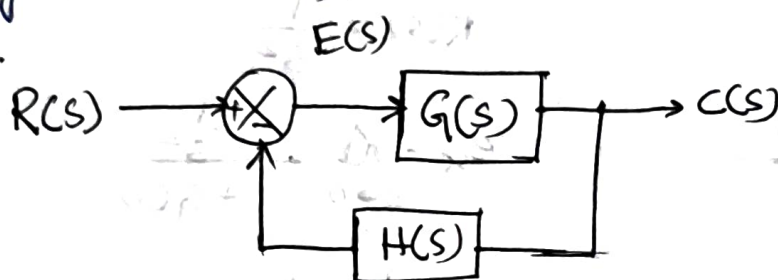
	Type-0	Type-1	Type-2
Step i/p	$\frac{1}{1+K}$	0	0
Ramp i/p	∞	$\frac{1}{K}$	0
Parabolic i/p	∞	∞	$\frac{1}{K}$

→ As the ~~steady~~ type no. (n) increases, steady state error reduces.

→ If we go beyond 2 (type no.), steady state error will reduce by dynamic error increases and difficult to stabilize.

→ Also we will not get information, how the steady state error varies with time.

* Steady state error for non-unity feedback control system



$$C(s) = \frac{G(s)}{1 + G(s)H(s)} R(s) \quad \text{--- (1)}$$

$$E(s) = R(s) - C(s)H(s)$$

$$\therefore C(s) = E(s)G(s)$$

$$\text{(1)} \Rightarrow E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

Steady state error,

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{R(s)}{1 + G(s)H(s)}$$

Q. For a unit feedback control system, the forward path transfer function is given by —

$$G(s) = \frac{20}{s(s+2)(s^2+2s+20)}, \text{ Determine the}$$

steady state error of the system, when the i/p are

- ① 5 ② $5t$ ③ $\frac{3t^2}{2}$

Soln:

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)}$$

① $R(s) = \frac{5}{s}$

② $R(s) = \frac{5}{s^2}$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{5}{s}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{\frac{5}{s}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= 0$$

$$= \lim_{s \rightarrow 0} \frac{5}{s + \frac{20}{(s+2)(s^2+2s+20)}}$$

③ $R(s) = \frac{3}{2s^3}$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{3}{2s^3}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \frac{5}{20} = \frac{5}{2}$$

$$= \lim_{s \rightarrow 0} \frac{\frac{3}{2s^2}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \rightarrow 0} \frac{\frac{3}{2}}{\frac{s^2 + 20s}{(s+2)(s^2+2s+20)}} = \infty$$

Q. A 2nd order sys. has transfer function given by - (GATE 1991)

$$G(s) = \frac{25}{s^2 + 8s + 25}, \text{ if the sys. initially at rest}$$

is subjected to a unit step input at $t=0$, the 2nd peak in the response will occur at ~~5 sec~~ (a) π sec (b) $\pi/2$ sec (c) $2\pi/3$ sec (d) $\pi/3$ sec

Soln: Stand. 2nd order sys. T.F. -

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\therefore \omega_n = 5 \text{ Rad/s}$$

$$2\zeta\omega_n = 8$$

$$\Rightarrow \zeta = \frac{8}{2 \times 5} = 0.8$$

$$\text{Now, } t_p = \frac{n\pi}{\omega_d} = \frac{n\pi}{\omega_n \sqrt{1-\zeta^2}}$$

For 2nd peak, $n=3$

$$t_p = \frac{3\pi}{5 \sqrt{1-(0.8)^2}} \text{ sec} = \pi \text{ sec}$$

Q. The open loop T.F. of a unity feedback sys. is given by -

$$G(s) = \frac{K}{s(1+st)}, \text{ where, } K \text{ and } T \text{ are positive const. By}$$

what factor should the amplifier gain be reduced so that the peak overshoot of unit step response of the sys. is reduced from 75% to 25%.

Soln:

$$\text{Closed loop TF. } \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{K}{s(1+sT)} = \frac{K}{s^2T + s + K} = \frac{K/T}{s^2 + \frac{1}{T}s + \frac{K}{T}}$$

$$\text{Here } \omega_n^2 = \frac{K}{T}$$

$$\Rightarrow \omega_n = \sqrt{\frac{K}{T}} \text{ rad/sec}$$

$$2\zeta\omega_n = \frac{1}{T}$$

$$\Rightarrow \zeta = \frac{1}{2T \times \sqrt{\frac{K}{T}}} = \frac{1}{2\sqrt{KT}}$$

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\text{Now, } e^{-\frac{\pi\zeta_1}{\sqrt{1-\zeta_1^2}}} = 0.75$$

$$e^{-\frac{\pi\zeta_2}{\sqrt{1-\zeta_2^2}}} = 0.25$$

$$\Rightarrow \zeta_1 = 0.09118$$

$$\Rightarrow \zeta_2 = 0.4037$$

$$\therefore \frac{\zeta_1}{\zeta_2} = \frac{\frac{1}{2\sqrt{KT}}}{\frac{1}{2\sqrt{K_2T}}} = \sqrt{\frac{K_2}{K}} = \frac{0.09118}{0.4037}$$

$$\boxed{K_1 = 19.6 K_2}$$

Stability:-

An LTI system is stable if the following two notions of stability are satisfied -

- ① If i/p is bounded, output is bounded (BIBO)
- ② If there is no i/p, o/p tends towards zero
- asymmetric stability.

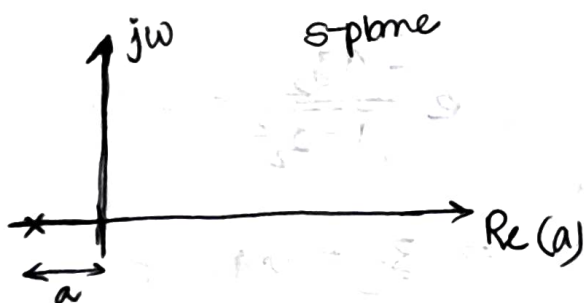
$$\frac{d^2 y}{dt^2} + 2 \zeta \omega_n \frac{dy}{dt} + \omega_n^2 y = x(t)$$

Unbounded i/p $\Rightarrow$ unbounded o/p.

Then nothing can be said about such sys.

depends on -
initial condⁿ
sys parameters
Small change in the
i/p large change in o/p.

Pole position:

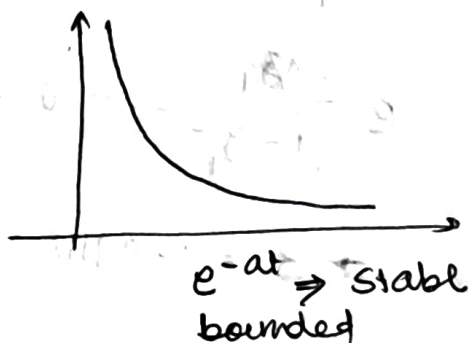


$$\frac{C(s)}{R(s)} = \frac{1}{s+a}$$

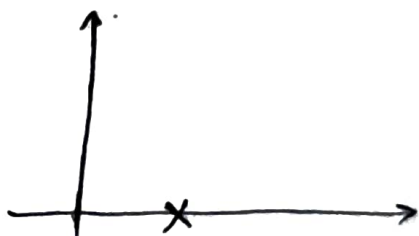
$$C(s) = \frac{1}{s+a}$$

$$\Rightarrow c(t) = e^{-at}$$

Impulse response

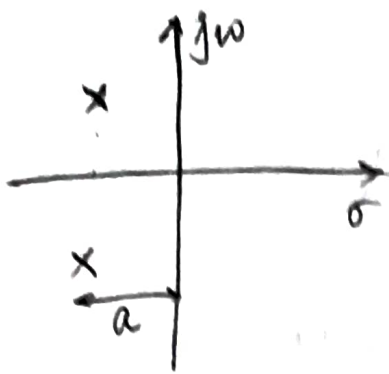


Pole position:



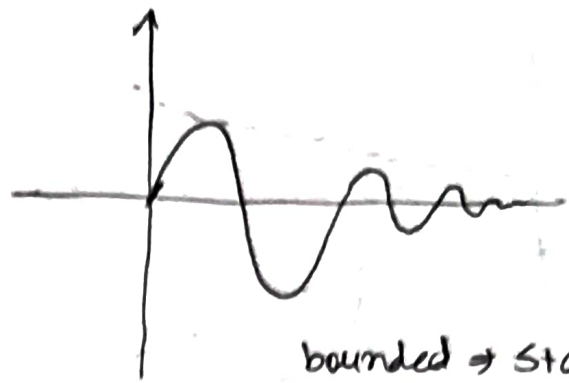
Impulse response





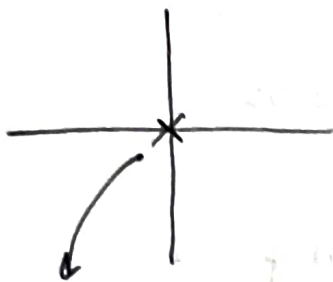
TF: $\frac{1}{(s+a)^2 + \omega_n^2}$

$C(t) = e^{-at} \sin \omega t$



bounded $\Rightarrow$ stable

$e^{-at} \sin \omega t$



TF = $\frac{1}{s}$

$C(s) = \frac{1}{s} R(s)$

$= \frac{1}{s} \cdot \frac{1}{s+1}$

$= \frac{1}{s} - \frac{1}{s+1}$

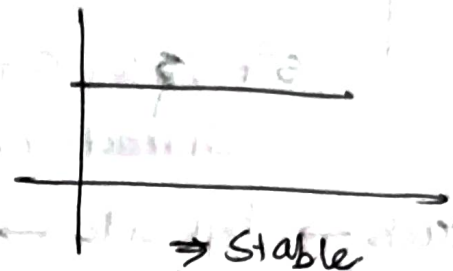
marginally stable;

root posⁿ same

for different $C(t) = (1 - e^{-t}) u(t)$

subjected signal

diff o/p; st or unit.

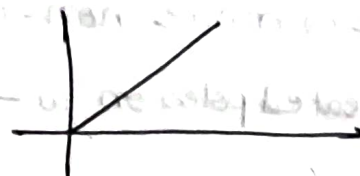


$\Rightarrow$ Stable

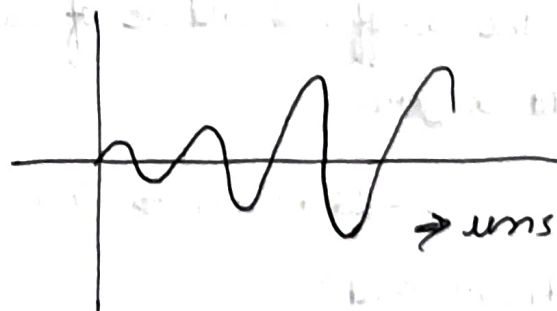
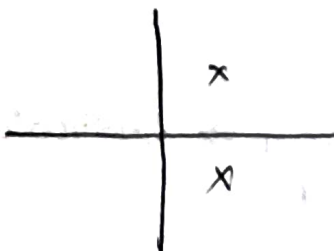
Now, subjected to unit step.

$C(s) = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$

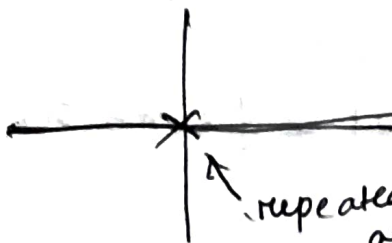
$C(t) = t u(t)$



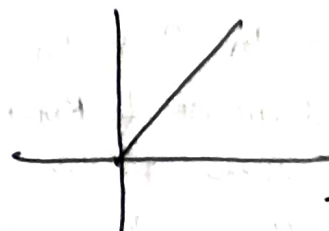
unstable.



$\Rightarrow$ unstable



repeated root at origin

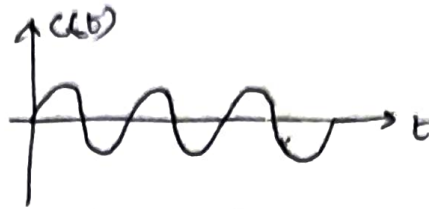


$\Rightarrow$ unstable

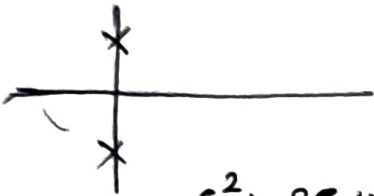


$$C(s) = \frac{1}{s^2 + \omega^2}$$

$$c(t) = \sin \omega t$$

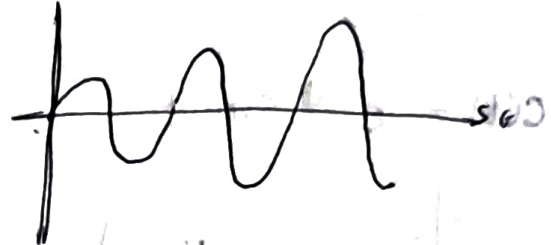


⇒ Stable
(marginally stable)



$$s^2 + 2\zeta \omega_n s + \omega_n^2 = 0$$

characteristic eqn.



⇒ Unstable

Roots → left side → stable

Roots → on $j\omega$ → single pair → marginally stable

Roots → on right → unstable.

Roots → on $j\omega$ (repeated) → unstable.

→ Conclusion: poles with negative real part stable system.

poles with positive real part unstable.

one or more non-repeated poles on $j\omega$ → marginally stable
or limited by stable

Repeated poles on $j\omega$ → unstable.

* Necessary conditions for stability:-

$$G(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0 = 0$$

all the co-eff. should be of same sign (real) no. co eff. should be zero.

$$s^2 - 2s + 1 = 0 \Rightarrow \text{co-eff. different sign} \Rightarrow \text{unstable.}$$

* Sufficient condⁿ:-

$$G(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0 = 0$$

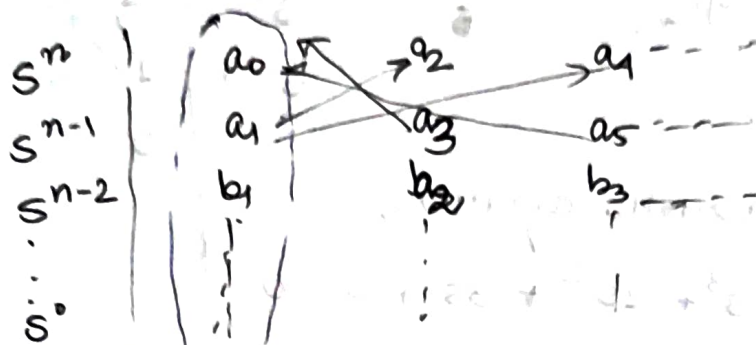
first column elements of Routh array should be of some sign for the sys. will be stable.

if any sign changes → unstable

no. of sign changes indicated no. of roots in the right half of s-plane.

* Routh stability criterion:-

$$G(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$



$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

first column
if same sign
→ stable.

+a₀
+a₁
-b₁
+b₂
} sign change = 2, means 2 roots in right side of s-plane.

Prob 1:- $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$

The Routh Table:

s^4	1	18	5
s^3	8	16	
s^2	$\frac{18-2}{1} = 16$	$\frac{5-0}{1} = 5$	
s^1	$27/16$	0	
s^0	5		

no sign change → stable

any row can be divided by common factor

Prob 2:- $3s^4 + 10s^3 + 5s^2 + 5s + 2 = 0$

s^4	3	5	2
s^3	10	5	0
s^2	$\frac{50-15}{10} = \frac{7}{2}$	2	
s^1	$-\frac{1}{7}$	0	
s^0	2	0	

$$\frac{\frac{7}{2} - 4}{7/2} = -\frac{1}{2} \times \frac{2}{7}$$

Here is a sign change → unstable
There are two sign changes → two roots in Right side. s-plane.

Special Cases:

- ① When the 1st term in any row is zero, while rest of the row have atleast one non-zero term. (It has to first fulfill the necessary condition)

$$q(s) = s^n + () s^{n-p} + () \dots$$

$$\begin{array}{c|ccc} s^n & 1 & 2 & \dots \\ s^{n-p} & 1 & 2 & \dots \\ & 0 & \dots & \dots \end{array}$$

eg:- Consider the characteristic equation,

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

The Routh array is -

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 3 \\ s^4 & 1 & 2 & 5 \\ s^3 & \frac{0(+)}{\epsilon} & -2 & \\ s^2 & \frac{2\epsilon+2(+)}{\epsilon} & 5 & \\ s^1 & \frac{-4\epsilon-4-5\epsilon^2}{2\epsilon+2} & (-) & \\ s^0 & 5 & (+) & \end{array}$$

NOTE: If any one of the element is zero, then, s^2 elements will be undefined.

Because of this zero term, in the next row, the terms becomes infinite and Routh test breakdown.

Method: Substitute a small positive no. ' ϵ ' for the zero, and proceed to evaluate the rest of the Routh array, then examine the sign of the first column of Routh array by making ϵ tends to zero.

there is two sign changes $(+ \rightarrow - \rightarrow +)$

$\therefore$ The system is unstable, With two poles in the right-half s-plane: